EIE3320 LAB 1 REPORT

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1. Introduction

This program is to display the image of the shapes of circle, rectangle and square. We used Bluej to complete the programming work.

In the program, user need to choose the shape (circle, rectangle or square) and input the details of the shape (radius, width or length) that the user wants to create. Then, the program will compute the area and perimeter of the shape and show the drawing of the shape.

1.1 Menu

The menu is to show the choices that the user can input.

```
**************************************
* Please choose one the followings:  *
* Press 'c' - Circle                 *
* Press 's' - Square                 *
* Press 'r' - Rectangle              *
* Press 'x' - EXIT                   *
**************************************
```

Figure 1.1: The menu

1.2 Circle

When the user enters “c”, that means the user is going to print the image of circle. If the user does not input an expected input, “Invalid Input” message will display and ask the user input again. The program will ask the user to input the radius of the circle and compute the area and perimeter of the circle.

```
***************************************
* Please choose one the followings:  *
* Press 'c' - Circle                 *
* Press 's' - Square                 *
* Press 'r' - Rectangle              *
* Press 'x' - EXIT                   *
***************************************
c
Please input the radius:
10
Area of circle = 314.15927
Perimeter of circle = 62.831856
***************************************
* Please choose one the followings:  *
* Press 'c' - Circle                 *
* Press 's' - Square                 *
* Press 'r' - Rectangle              *
* Press 'x' - EXIT                   *
***************************************
```

Figure 1.2.1: result of the input circle

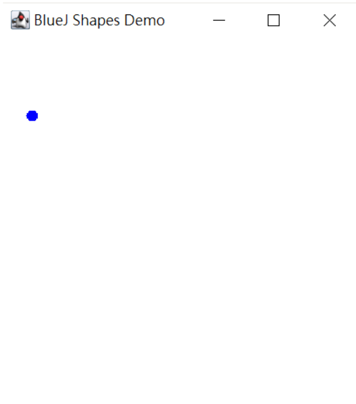


Figure 1.2.2: image of the circle

1.3 Square

When the user enters “s”, that means the user is going to print the image of Square. If the user does not input an expected input, “Invalid Input” message will display and ask the user input again. The program will ask the user to input the length of the side of the square and compute the area and perimeter of the square.

```
***************************************
* Please choose one the followings:  *
* Press 'c' - Circle                 *
* Press 's' - Square                 *
* Press 'r' - Rectangle              *
* Press 'x' - EXIT                   *
***************************************
s
Please input the length:
40
Area of square = 1600.0
Perimeter of square = 160.0
***************************************
* Please choose one the followings:  *
* Press 'c' - Circle                 *
* Press 's' - Square                 *
* Press 'r' - Rectangle              *
* Press 'x' - EXIT                   *
***************************************
```

Figure 1.3.1: result of the input of square

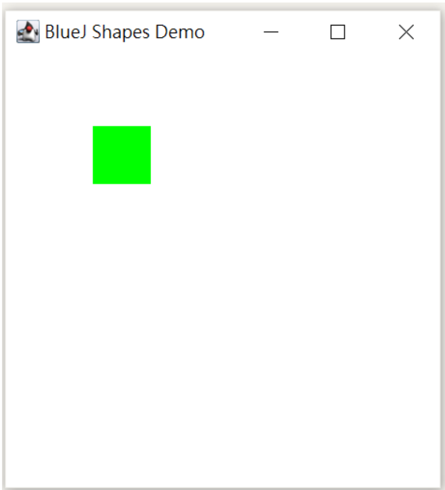


Figure 1.3.2: image of the square

1.4 Rectangle

When the user enters “r”, that means the user is going to print the image of rectangle. If the user does not input an expected input, “Invalid Input” message will display and ask the user input again. The program will ask the user to input the length and the width of the rectangle and compute the area and perimeter of the rectangle.

```
**************************************
* Please choose one the followings:  *
* Press 'c' - Circle                 *
* Press 's' - Square                 *
* Press 'r' - Rectangle              *
* Press 'x' - EXIT                   *
**************************************
r
Please input the length:
10
Please input the width:
20
Area of rectangle = 200.0
Perimeter of rectangle = 60.0
**************************************
* Please choose one the followings:  *
* Press 'c' - Circle                 *
* Press 's' - Square                 *
* Press 'r' - Rectangle              *
* Press 'x' - EXIT                   *
**************************************
```

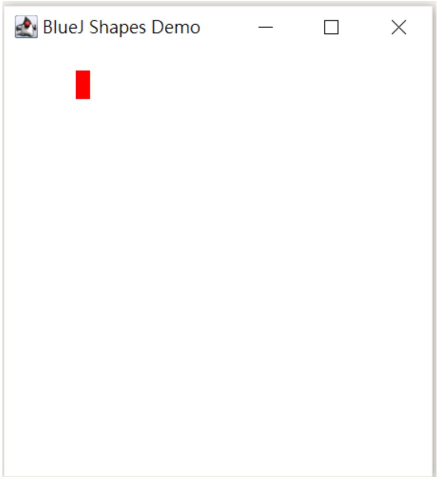


Figure 1.4.1: result of the input of rectangle

Figure 1.4.2: image of the rectangle

1.5 Exit

When the user enters “x”, the program will be ended. If the user does not input an expected input, “Invalid Input” message will display and ask the user input again.

2. Methodology

2.1 Division of work

In this project, Law is better in programming and So is better of report writing. Therefore, Law has focused on building the program and So has focused on debugging and report writing.

2.2 Schedule

Date	Content
22-23/10/2020	Programming
24/10/2020	Testing and debugging
25-26/10/2020	Report writing

2.3 Program structure

2.3.1 Class “Shape”

“Shape” is a superclass of the subclasses “Circle, “Square” and “Rectangle”. It contains 2 protected data members and 4 abstract methods.

Shape
-area: float // to store the area of the shape -perimeter: float // to store the perimeter of the shape
+readShape(): void // to read the input from the user +computeArea(): void // to compute the area of the shape with the user’s input +computePerimeter(): void // to compute the perimeter of the shape with the user’s input +displayShape(): to display the area and perimeter of the shape

“Shape” is an abstract, so the methods can override to the subclasses.

2.3.2 Interface Class “Drawable”

The interface class Drawable is used to display the images of the shapes, which is draw by the method “Canvas”.

Drawable
+draw(): void

2.3.3 Class “Canvas”

This class is used to draw the images of the shapes.

2.3.4 SubClass “Circle”

“Circle” is a subclass that contains a private data member and 7 methods. As the members, “area” and “perimeter”, have been created in the superclass, there is not necessary to create in the subclass.

Circle
-radius: float // to store the radius of the circle
-area: float // to store the area of the circle (has been created in the superclass) -perimeter // to store the perimeter of the circle (has been created in the superclass)

+Circle() // to create an object in class Circle
+Circle(r: float) // to create an object in class Circle with r
+readShape(): void // to read the radius input from the user and set it as radius
+computeArea(): void // to compute the area by radius*radius*Math.PI
+computePerimeter(): void // to compute the perimeter by 2 *Math.PI*radius
+displayShape(): void // to show the computed area and perimeter
+draw(): void // to draw the Circle based on the calculation

2.3.5 SubClass “Square”

“Square” is a subclass that contains a private data member and 7 methods. As the members, “area” and “perimeter”, have been created in the superclass, there is not necessary to create in the subclass.

Square
-length: float // to store the length of the square
-area: float // to store the area of the square (has been created in the superclass) -perimeter // to store the perimeter of the square (has been created in the superclass)

+ Square() // to create an object in class Square
+ Square(l: float) // to create an object in class Square with l
+readShape(): void // to read the length input from the user and set it as length
+computeArea(): void // to compute the area by length*length
+computePerimeter(): void // to compute the perimeter by 4*length
+displayShape(): void // to show the computed area and perimeter
+draw(): void // to draw the Square based on the calculation

2.3.6 SubClass “Rectangle”

“Rectangle” is a subclass that contains 2 private data members and 7 methods. As the members, “area” and “perimeter”, have been created in the superclass, there is not necessary to create in the subclass.

Rectangle
-length: float // to store the length of the rectangle -width: float // to store the width of the rectangle
-area: float // to store the area of the rectangle (has been created in the superclass) -perimeter // to store the perimeter of the rectangle (has been created in the superclass)

+ Rectangle() // to create an object in class Rectangle
+ Rectangle (l: float, w: float) // to create an object in class Rectangle with l and w
+readShape(): void // to read the length and width input from the user and set it as length and width
+computeArea(): void // to compute the area by length*width
+computePerimeter(): void // to compute the perimeter by (length*2)+(width*2)
+displayShape(): void // to show the computed area and perimeter
+draw(): void // to draw the Rectangle based on the calculation

2.3.7 Class “PictureTester” and “ShaperTester”

These two classes are used to test the program wheather the program run correct or not.

2.3.8 Class ”Picture”

“Picture” is a class using an array to store the shape createed by the user.

Picture
-shapes: ArrayList<Shape> // to store the data of the shapes that the user want to create
+Picture() // create an object in class Picture +addShape(s:Shape): void // add a new shape in the arraylist +computeShape(): void // to compute the area and perimeter of the shape +listAllShapeTypes(): void // list put all the sjapes in the arraylist +listSingShapeType(className: String): void // list out all shapes in the targeted type

3. Program Testing

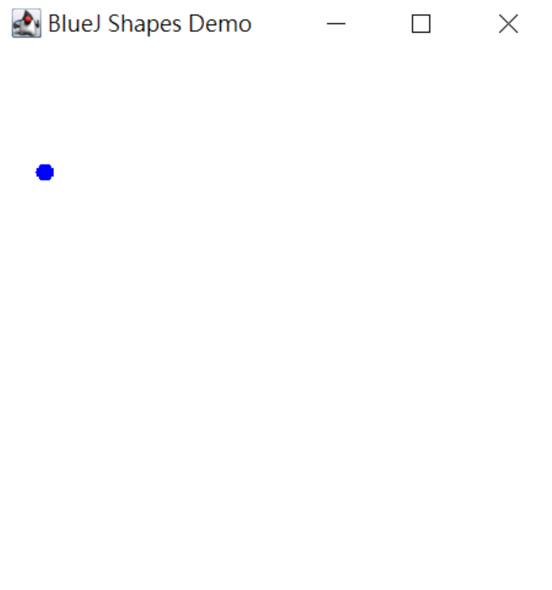
To test our program, we trtes different cases for our program and check the result if the output is as expected or not.

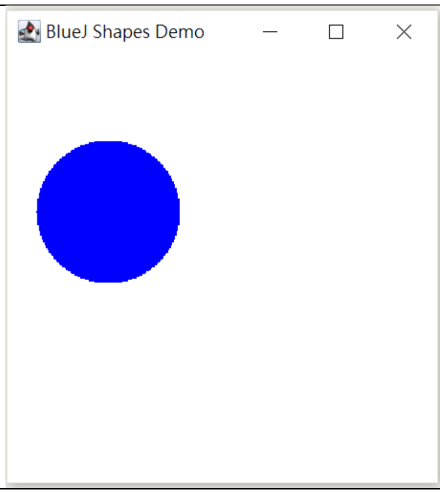
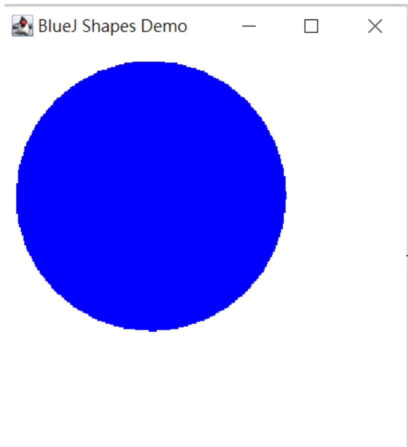
3.1 Menu

Input	Expected Result	Actual result
'c'	Ask user input the data of circle	Same as expected ** * Please choose one the followings: * * Press 'c' - Circle * * Press 's' - Square * * Press 'r' - Rectangle * * Press 'x' - EXIT * ** c Please input the radius:
's'	Ask user input the data of square	Same as expected ** * Please choose one the followings: * * Press 'c' - Circle * * Press 's' - Square * * Press 'r' - Rectangle * * Press 'x' - EXIT * ** s Please input the length:
'r'	Ask user input the data of rectangle	Same as expected ** * Please choose one the followings: * * Press 'c' - Circle * * Press 's' - Square * * Press 'r' - Rectangle * * Press 'x' - EXIT * ** r Please input the length:
'x'	End program	Same as expected ** * Please choose one the followings: * * Press 'c' - Circle * * Press 's' - Square * * Press 'r' - Rectangle * * Press 'x' - EXIT * ** x Can only enter input while your programm
'abc'	Display "Invalid Input" message and ask the user input again	Same as expected ** * Please choose one the followings: * * Press 'c' - Circle * * Press 's' - Square * * Press 'r' - Rectangle * * Press 'x' - EXIT * ** abc Invalid command!

3.2 Circle

Inputting radius:

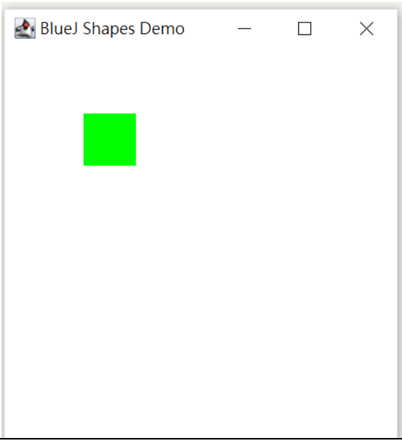
Input	**Expected Result**	**Actual result**
10	Display the area, perimeter and the image of the circle	Same as expected ************************************** * Please choose one the followings: * * Press 'c' - Circle * * Press 's' - Square * * Press 'r' - Rectangle * * Press 'x' - EXIT * ************************************** c Please input the radius: 10 Area of circle = 314.15927 Perimeter of circle = 62.831856 
100	Display the area, perimeter and the image of the circle	Same as expected ************************************** * Please choose one the followings: * * Press 'c' - Circle * * Press 's' - Square * * Press 'r' - Rectangle * * Press 'x' - EXIT * ************************************** c Please input the radius: 100 Area of circle = 31415.928 Perimeter of circle = 628.31854

<table>
<tr><td></td><td></td><td>
</td></tr>
<tr><td>100.1</td><td>Display the area, perimeter and the image of the circle</td><td>Same as expected
****
* Please choose one the followings: *
* Press 'c' - Circle *
* Press 's' - Square *
* Press 'r' - Rectangle *
* Press 'x' - EXIT *
****
c
Please input the radius:
100.1
Area of circle = 31478.79
Perimeter of circle = 628.94684

</td></tr>
<tr><td>EIE3320</td><td>Runtime error</td><td>Same as expected
****
* Please choose one the followings: *
* Press 'c' - Circle *
* Press 's' - Square *
* Press 'r' - Rectangle *
* Press 'x' - EXIT *
****
c
Please input the radius:
EIE3320
Can only enter input while your programming is running
java.util.InputMismatchException
at java.base/java.util.Scanner.throwFor(Scanner.ja
at java.base/java.util.Scanner.next(Scanner.java:1
at java.base/java.util.Scanner.nextFloat(Scanner.j
at Circle.readShape(Circle.java:36)
at ShapeTester.main(ShapeTester.java:28)</td></tr>
</table>

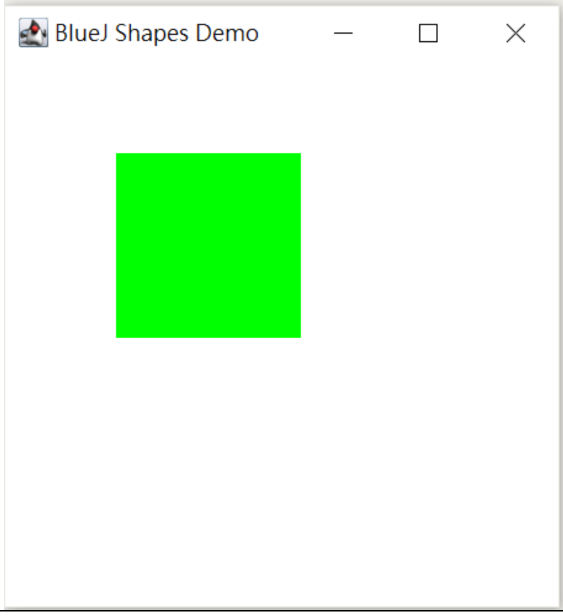
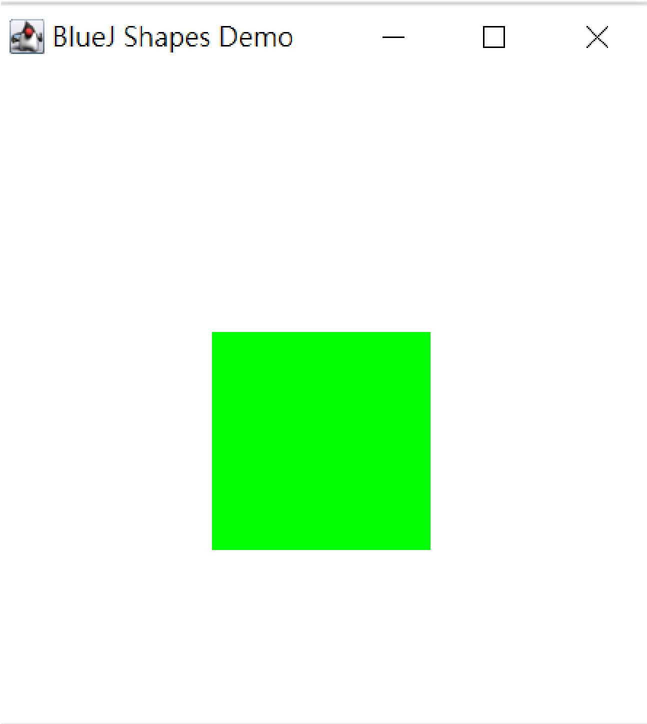
3.3 Square

Inputting length:

Input	Expected Result	Actual result

40	Display the area, perimeter and the image of the square	Same as expected
		`**************************************`
		`* Please choose one the followings:  *`
		`* Press 'c' - Circle                 *`
		`* Press 's' - Square                 *`
		`* Press 'r' - Rectangle              *`
		`* Press 'x' - EXIT                   *`
		`**************************************`
		`s`
		`Please input the length:`
		`40`
		`Area of square = 1600.0`
		`Perimeter of square = 160.0`
		

100	Display the area, perimeter and the image of the square	Same as expected
		`**************************************`
		`* Please choose one the followings:  *`
		`* Press 'c' - Circle                 *`
		`* Press 's' - Square                 *`
		`* Press 'r' - Rectangle              *`
		`* Press 'x' - EXIT                   *`
		`**************************************`
		`s`
		`Please input the length:`
		`100`
		`Area of Square = 10000.0`
		`Perimeter of Square = 400.0`

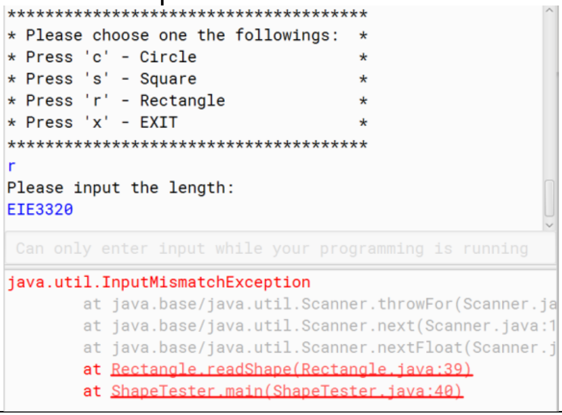
<table>
<tr><td></td><td></td><td>
</td></tr>
<tr><td>100.1</td><td>Display the area, perimeter and the image of the square</td><td>Same as expected
<pre>
****
* Please choose one the followings: *
* Press 'c' - Circle   *
* Press 's' - Square   *
* Press 'r' - Rectangle   *
* Press 'x' - EXIT   *
****
s
Please input the length:
100.1
Area of Square = 10020.01
Perimeter of Square = 400.4
</pre>

</td></tr>
</table>

<table>
<tr><td>EIE3320</td><td>Runtime error</td><td>Same as expected
****
* Please choose one the followings: *
* Press 'c' - Circle   *
* Press 's' - Square   *
* Press 'r' - Rectangle   *
* Press 'x' - EXIT   *
****
s
Please input the length:
EIE3320
Can only enter input while your programming is running
java.util.InputMismatchException
at java.base/java.util.Scanner.throwFor(Scanner.ja
at java.base/java.util.Scanner.next(Scanner.java:1
at java.base/java.util.Scanner.nextFloat(Scanner.j
at Square.readShape(Square.java:36)
at ShapeTester.main(ShapeTester.java:34)</td></tr>
</table>

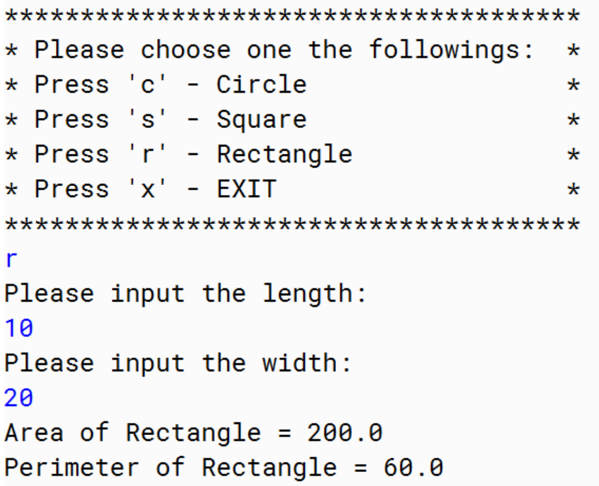
3.4 Rectangle

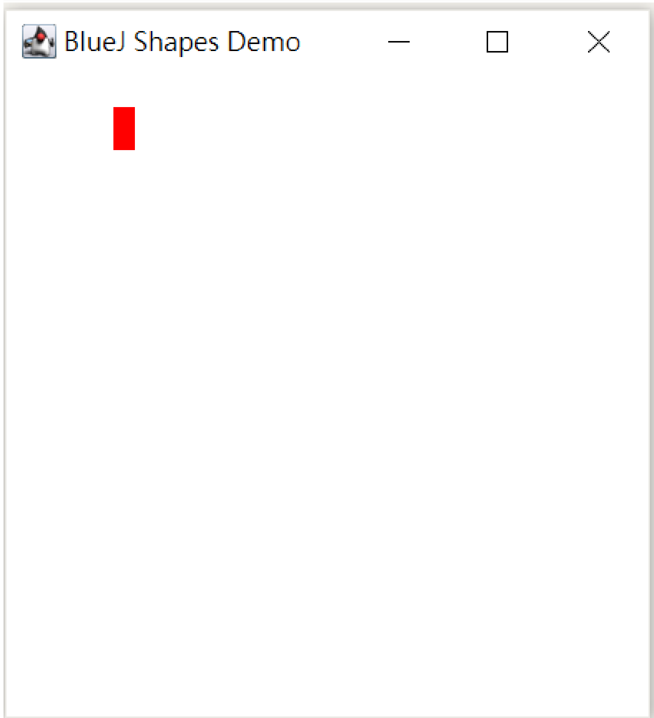
Inputting length :

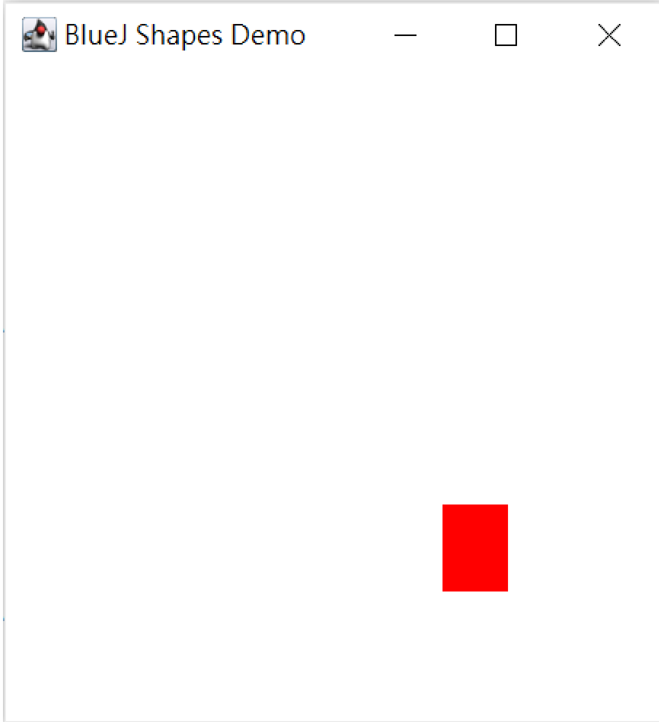
<table>
<tr><th>Input</th><th>Expected Result</th><th>Actual result</th></tr>
<tr><td>10</td><td>Ask the user to input the width</td><td>Same as expected
****
* Please choose one the followings: *
* Press 'c' - Circle   *
* Press 's' - Square   *
* Press 'r' - Rectangle   *
* Press 'x' - EXIT   *
****
r
Please input the length:
10
Please input the width:</td></tr>
<tr><td>30</td><td>Ask the user to input the width</td><td>Same as expected
****
* Please choose one the followings: *
* Press 'c' - Circle   *
* Press 's' - Square   *
* Press 'r' - Rectangle   *
* Press 'x' - EXIT   *
****
r
Please input the length:
30
Please input the width:</td></tr>
<tr><td>50.1</td><td>Ask the user to input the width</td><td>Same as expected
****
* Please choose one the followings: *
* Press 'c' - Circle   *
* Press 's' - Square   *
* Press 'r' - Rectangle   *
* Press 'x' - EXIT   *
****
r
Please input the length:
50.1
Please input the width:</td></tr>
</table>

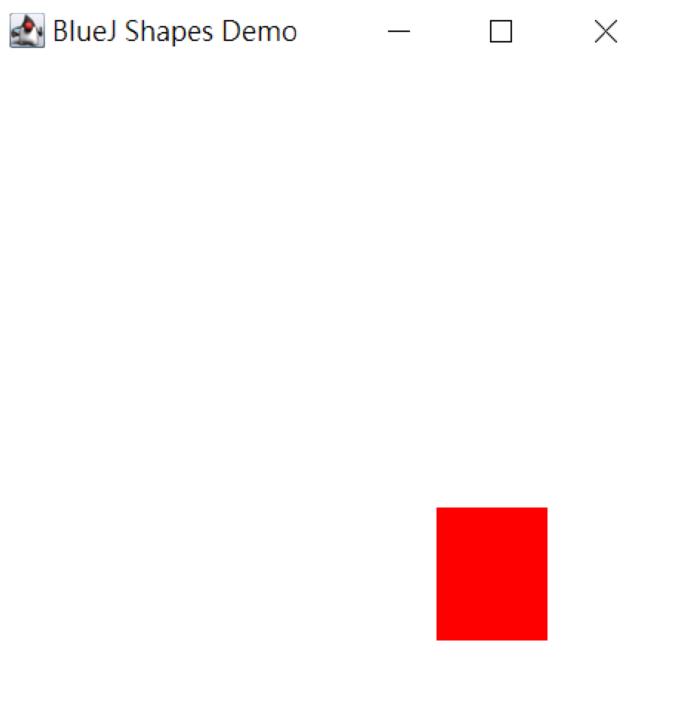
<table>
<tr><td>EIE3320</td><td>Runtime error</td><td>Same as expected

</td></tr>
</table>

Inputting width :

<table>
<tr><th>Input</th><th>Expected Result</th><th>Actual result</th></tr>
<tr><td>20</td><td>Display the area, perimeter and the image of the rectangle</td><td>Same as expected



</td></tr>
</table>

40	Display the area, perimeter and the image of the rectangle	Same as expected ************************************** * Please choose one the followings: * * Press 'c' - Circle * * Press 's' - Square * * Press 'r' - Rectangle * * Press 'x' - EXIT * ************************************** r Please input the length: 30 Please input the width: 40 Area of Rectangle = 1200.0 Perimeter of Rectangle = 140.0 
60.1	Display the area, perimeter and the image of the rectangle	Same as expected ************************************** * Please choose one the followings: * * Press 'c' - Circle * * Press 's' - Square * * Press 'r' - Rectangle * * Press 'x' - EXIT * ************************************** r Please input the length: 50.1 Please input the width: 60.1 Area of Rectangle = 3011.0098 Perimeter of Rectangle = 220.4

<table>
<tr><td></td><td></td><td>
</td></tr>
<tr><td>EIE3320</td><td>Runtime error</td><td>Same as expected<pre>
****
* Please choose one the followings: *
* Press 'c' - Circle   *
* Press 's' - Square   *
* Press 'r' - Rectangle   *
* Press 'x' - EXIT   *
****
r
Please input the length:
1
Please input the width:
EIE3320
Can only enter input while your programming is running
java.util.InputMismatchException
	at java.base/java.util.Scanner.throwFor(Scanner.java:939
	at java.base/java.util.Scanner.next(Scanner.java:1594)
	at java.base/java.util.Scanner.nextFloat(Scanner.java:24
	at Rectangle.readShape(Rectangle.java:41)
	at ShapeTester.main(ShapeTester.java:40)
</pre></td></tr>
</table>

4. Conclusion

In this exercise, we can know more about how to use abstract class, superclass, and subclass. We have deeply understand the use of these classes and how we connect the classes, instead of typing a long program in the main class.

When we are working on this exercise, we found there are some interesting bugs. For example, we display a circle with an angle, and we can overlap all images with bugs to form an interest shape. Also, we had a problem on drawing the image using the class “Canvas” that we have to quick scan the whole program.

We both agree that this exercise is not that diffcult to finish, comparing to the Homeworks. This exercise requires more programming and logic skills when we build the whole program. And this exercise is more complicated that lots of classes and methods are used. But they are staright forward that it is easy to understand when we completed the whold program.

5. Future Development

5.1 The limit of input values

The radius, length and width can be input as a negative number or '0' and the program would still compute the result, which not possible. Also, as radius, length and width are float, which can only store 32 bits value. Therefor, the value can be used as double to store 64 bits value.
Also, the length and width of the rectangle can be input as two same values, which is a square, not a rectangle. The program should check if the length and width of the rectangle should not be the same.

5.2 GUI

This program can be more user-friendly that GUI (graphical user interface) cab be used. User can choose their choices on choosing the shape that te user what to create by button, instead of typing, to reduce the chance of typo.

5.3 Colour of the shape

In the program, it can only display objects with singal color. Therfore, the program should allow the user to choose the color of the shape. A menu of color can be added after the use input the data of the shape.

5.4 Display more shapes

This program is limted on displaying circle, square and rectangle. More shapes can be added, e.g. triangle, trapezium, not only the current shapes.